

# TAKAHIRO YOSHIDA's WEBSITE

June, 2026



# Contents

|                                       |           |
|---------------------------------------|-----------|
| <b>Welcome!</b>                       | <b>5</b>  |
| Updates . . . . .                     | 5         |
| Educational background . . . . .      | 7         |
| Academic experience . . . . .         | 7         |
| Contact . . . . .                     | 8         |
| <b>Publications</b>                   | <b>9</b>  |
| In progress / under review . . . . .  | 9         |
| Refereed journal papers . . . . .     | 9         |
| Refereed proceedings papers . . . . . | 15        |
| Book chapters / sections . . . . .    | 24        |
| Book translation . . . . .            | 26        |
| <b>Honors</b>                         | <b>27</b> |
| Awards . . . . .                      | 27        |
| Research grants . . . . .             | 28        |
| <b>Notes: R and Python</b>            | <b>31</b> |



# Welcome!

**Takahiro Yoshida** is an Associate Professor at Center for Spatial Information Science, the University of Tokyo, Japan. The foundation of all his research interests is **Geographical Information Science**. His more specific interests are as follows:

- (1) **Urban and Regional Analysis** based on spatial statistics, spatial econometrics, remote sensing, and its combination with compositional data analysis,
- (2) **Climate Change and Cities** with complex systems of building environments, transport networks, and human behaviors.

researchmap

## Updates

- **June 8, 2026 [Event]** hosted International Workshop on Urban Analytics 2026 with Georgia Institute of Technology (June 8–20, 2026)
- **June 3, 2026 [Conference]** presented a leading co-authored paper in the 11th International Workshop on Compositional Data Analysis (CoDa-Work2026)
- **May 31, 2026 [Event]** visited the Bartlett Centre for Advanced Spatial Analysis, University College London
- **May 25, 2026 [Publication]** accepted a co-authored paper in *The Lancet Regional Health - Western Pacific*
- **May 19, 2026 [Conference]** presented a co-authored paper in the 12th International Conference on Indoor Air Quality, Ventilation and Energy Conservation in Buildings (IAQVEC2026)
- **April 1, 2026 [Event]** appointed as Visiting Associate Professor at the Institute of Statistical Mathematics
- **April 1, 2026 [Event]** promoted to Associate Professor at CSIS
- **February 11, 2026 [Publication]** accepted a co-authored paper in *Geographical Analysis*

- **February 6, 2026 [Event]** served as a guest lecturer and a Ph.D. defense committee member at GeorgiaTech
- **December 5, 2025 [Publication]** accepted a co-authored paper in *Sustainability Science*
- **November 2, 2025 [Conference]** presented a leading poster and a co-authored poster, and co-hosted a special session The Future of Spatial Information Science: Discovered by Novel Spatial Data in the 34th annual meeting of the Geographic Information Systems Association (GISA2025)
- **October 28, 2025 [Event]** presented at Urban AI Symposium with Center for Urban Science + Progress, Tandon School of Engineering, New York University
- **October 27, 2025 [Publication]** accepted a co-authored paper in *Journal of Transport & Health*
- **September 7, 2025 [Publication]** published a lead authored book section in *Urban Planning Towards Carbon Neutrality* (in Japanese)
- **September 2, 2025 [Conference]** presented one co-authored paper and one co-authored poster in the 20th International 3D GeoInfo Conference and the 9th International Smart Data and Smart Cities Conference (3D GeoInfo & SDSC 2025)
- **August 30, 2025 [Publication]** accepted a co-authored paper in *Applied Energy*
- **August 18, 2025 [Event]** conducted the FY2026 Entrance Examination of Department of Socio-Cultural Environmental Studies, Graduate School of Frontier Sciences
- **August 10, 2025 [Publication]** published a lead authored book-section in *Thematic Encyclopedia of Regional Science*
- **July 18, 2025 [Conference]** presented one lead authored and two co-authored papers in the 11th Applied Energy Symposium and Forum: Low Carbon Cities and Urban Energy Systems (CUE2025)
- **July 15, 2025 [Conference]** presented a co-authored poster in the 7th Spatial Statistics Conference
- **July 7, 2025 [Conference]** presented a co-authored paper in the 12th International Conference on Urban Climate (ICUC12)
- **June 27, 2025 [Grant]** accepted and started a government funding as PI “Urban carbon mapping based on smart mater data”
- **June 23, 2025 [Conference]** presented a co-authored poster in the 19th International Conference on Computational Urban Planning and Urban Management (CUMPUM2025)
- **June 18, 2025 [Event]** hosted Cooperation Agreement Ceremony between CSIS, the University of Tokyo and School of City and Regional Planning, College of Design, Georgia Institute of Technology
- **June 9, 2025 [Event]** hosted International Workshop on Urban Analytics 2025 with Georgia Institute of Technology (June 9–22, 2025)
- **May 4, 2025 [Publication]** accepted a co-authored paper in *Applied Energy*
- **April 1, 2025 [Grant]** started the KAKENHI Grants-in-Aid for Scien-

tific Research (B) as PI “Spatial compositional data analysis”

## Educational background

- Ph.D. in Policy and Planning Sciences, University of Tsukuba, March 2018.
  - [Dissertation title] Methodological developments and socio-economic applications of compositional data analysis for geographical data.
- Master of Science in Engineering, University of Tsukuba, March 2015.
- Bachelor of Policy and Planning Sciences, University of Tsukuba, March 2013.

## Academic experience

- April 2026 – present
  - Associate Professor, Division of Spatial Information Analysis, Center for Spatial Information Science, the University of Tokyo, Japan.
- April 2024 – March 2026
  - Assistant Professor (Lecturer), Division of Spatial Information Analysis, Center for Spatial Information Science, the University of Tokyo, Japan.
- September 2022 – March 2024
  - Assistant Professor, Division of Spatial Information Analysis, Center for Spatial Information Science, the University of Tokyo, Japan.
- April 2021 – August 2022
  - Project Assistant Professor, Department of Urban Engineering, School of Engineering, the University of Tokyo, Japan.
- April 2018 – March 2021
  - Research Associate, Center for Global Environmental Research, National Institute for Environmental Studies, Japan.
- April 2016 – March 2018
  - Research Fellow (DC2), Japan Society for the Promotion of Science, Japan.

(Concurrent appointments in the university)

- April 2024 – present
  - Department of Socio-Cultural Environmental Studies, Graduate School of Frontier Sciences, the University of Tokyo, Japan.
- April 2024 – present
  - Collaborative Research Organization for the Digital Spatial Society, the University of Tokyo, Japan.
- April 2023 – present
  - Collaborative Research Organization for Future Regional Society, the University of Tokyo, Japan.

(Concurrent appointments out of the university)

- April 2026 – present
  - Visiting Associate Professor, Risk Analysis Research Center, the Institute of Statistical Mathematics, Japan.
- April 2023 – present
  - Visiting Researcher, Social System Division, National Institute for Environmental Studies, Japan.
- April 2021 – present
  - Part-Time Lecturer, Graduate School of System Design and Management, Keio University, Japan.
- April 2021 – March 2023
  - Visiting Researcher, Earth System Division, National Institute for Environmental Studies, Japan.
- April 2019 – March 2022
  - Part-Time Lecturer, Department of International Economics, Faculty of Economics, Toyo University, Japan.
- September 2013 – September 2013
  - Intern, Geographic Information Analysis Research Division, Geography and Crustal Dynamics Research Center, Geospatial Information Authority of Japan, Japan.

## Contact

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# Publications

## In progress / under review

- Takahiro Yoshida, Daisuke Murakami, Hajime Seya **Location powered quotient: A compositional data analysis-based approach.**
- Hajime Seya, Takahiro Yoshida **Propensity score matching for multiple treatment levels: A CODA-based contribution.** arXiv: 1710.08558 [LINK]
- Takahiro Yoshida, Daisuke Murakami, Hajime Seya, Morito Tsutsumi **A Monte Carlo study: Spatial multinomial discrete choice models.**
- Takahiro Yoshida, Daisuke Murakami, Hajime Seya, Morito Tsutsumi **Spatial seemingly unrelated regression models combined with compositional data analysis approach.**
- Takahiro Yoshida, Masato Tomonari, Hajime Seya, Morito Tsutsumi **On the relationships between amount of zero responses and goodness-of-fits of compositional models.**
- Takahiro Yoshida, Morito Tsutsumi, Hozana Ishii **Identification of regional difference of election voting trends: Functional compositional data clustering approach.**
- + four papers are under review.

## Refereed journal papers

1. Kazuaki Arai, Masamitsu Kamada, Takahiro Yoshida, Ikuho Yamada, Ding Ding, Kimihiro Hino (2026) **Cross-level regional and socioeconomic inequalities in step counts: A descriptive epidemiological investigation of 1.5 million Japanese smartphone users.** *The Lancet Regional Health - Western Pacific*, (published online first; accepted on May 24, 2026). DOI: 10.1016/j.lanwpc.2026.101901 [LINK] [PRESS RELEASE]
2. Daisuke Murakami, Alexis Comber, Takahiro Yoshida, Narumasa Tsutsumida, Chris Brunsdon, Tomoki Nakaya (2026) **Coarse-to-fine spatial**

- modeling: A scalable, machine-learning-compatible framework.** *Geographical Analysis*, 58 (2), e70034. DOI: 10.1111/gean.70034 [LINK]
3. Yasuo Takahashi, Takahiro Yoshida, Brian Alan Johnson, Sawako Shigeto, Yoshiki Yamagata (2026) **Contribution of natural capital to quality of life of urban-to-rural migrants in Japan: structural equation modelling using nation-wide survey and public geospatial data.** *Sustainability Science*, 21, 1157–1176. DOI: 10.1007/s11625-025-01788-3 [LINK]
  4. Kimihiro Hino, Ikuho Yamada, Takahiro Yoshida (2025) **The impact of introducing new light rail on step counts: Threshold distance analysis in a Japanese regional city.** *Journal of Transport & Health*, 45, 102206. DOI: 10.1016/j.jth.2025.102206 [LINK]
  5. Ryuta Tsurumi, Takahiro Yoshida (2025) **Impact of people flows on energy consumption in an office building in Tokyo using a state-space model.** *Applied Energy*, 401, 126697. DOI: 10.1016/j.apenergy.2025.126697 [LINK]
  6. Nguyen Thi Quynh Trang, Koharu Okada, Yusei Sugiyama, Takahiro Yoshida, Yujiro Hirano, Peraphan Jittrapirom, Tosiya Nakaegawa, Takuro Kobashi (2025) **On the decarbonization potentials of rooftop PVs integrated with EVs as battery for all the municipalities of Japan.** *Applied Energy*, 393, 126067. DOI: 10.1016/j.apenergy.2025.126067 [LINK]
  7. Liang Zhang, Hasi Bagan, Chaomin Chen, Takahiro Yoshida (2025) **Exploring the impact of urban morphology on river cooling effects: A case study of the Arakawa River in Tokyo.** *Ecological Indicators*, 172, 113288. DOI: 10.1016/j.ecolind.2025.113288 [LINK]
  8. Yujiro Hirano, Yukiko Yoshida, Takahiro Yoshida, Yoshiki Yamagata, Suguru Mizutani, Ji Xuan (2025) **Evaluation of CO2 emissions reduction in the Japanese residential sector through energy-saving scenarios based on large-scale survey data.** *Energies*, 18 (8), 1964. DOI: 10.3390/en18081964 [LINK]
  9. Ishwar D. Ramnarine, Tarek A. Sherif, Abdulrahman H. Alorabi, Haya Helmy, Takahiro Yoshida, Akito Murayama, Perry P. J. Yang (2025) **Urban revitalization pathways toward zero carbon emissions through systems architecting of urban digital twins.** *Environment and Planning B: Urban Analytics and City Science*, 52 (8), 1920–1948. DOI: 10.1177/23998083251318142 [LINK]
  10. Alexis Comber, Paul Harris, Daisuke Murakami, Tomoki Nakaya, Narumasa Tsutsumida, Takahiro Yoshida, Chris Brunsdon (2024) **Encapsulating spatially varying relationships with a generalized additive model.** *ISPRS International Journal of Geo-Information*, 13 (12), 459. DOI: 10.3390/ijgi13120459 [LINK]
  11. Terigelehu Te, Chunling Bao, Hasi Bagan, Yuxin Xie, Meihui Che, Takahiro Yoshida, Bayarsaikhan Uudus (2024) **Mapping seamless monthly XCO2 in East Asia: Utilizing OCO-2 data and machine learning.** *International Journal of Applied Earth Observation and*

- Geoinformation*, 133, 104117. DOI: 10.1016/j.jag.2024.104117 [LINK]
12. Junya Yamasaki, Yasutaka Wakazuki, Satoru Iizuka, Takahiro Yoshida, Ryoichi Nitantai, Rikutarō Manabe, Akito Murayama (2024) **Microclimate simulation for future urban district under SSP/RCP: Reflecting changes in building stocks and temperature rises.** *Urban Climate*, 57, 102068. DOI: 10.1016/j.uclim.2024.102068 [LINK]
  13. Luwen Tan, Yuxin Xie, Chaomin Chen, Hasi Bagan, Takahiro Yoshida (2024) **Urbanization and land subsidence: Multi-decadal investigation combined SBAS-InSAR and multi-factors in Shanghai, China.** *Geocarto International*, 39 (1), 2391056. DOI: 10.1080/10106049.2024.2391056 [LINK]
  14. Yasuo Takahashi, Takahiro Yoshida, Sawako Shigeto, Hiroyuki Kubota, Brian Johnson, Yoshiaki Yamagata (2024) **Spatial exploration of rural capital contributing to quality of life and urban-to-rural migration decisions: A case study of Hokuto City, Japan.** *Sustainability Science*, 19, 489–506. DOI: 10.1007/s11625-023-01427-9 [LINK]
  15. Takahiro Yoshida, Daisuke Murakami, Hajime Seya (2024) **Spatial prediction of apartment rent using regression-based and machine learning-based approaches with a large dataset.** *The Journal of Real Estate Finance and Economics*, 69 (1), 1–28. DOI: 10.1007/s11146-022-09929-6 [LINK] Note: Published online: November 2022; Issued with volume and pages numbers: July 2024.
  16. Daisuke Murakami, Narumasa Tsutsumida, Takahiro Yoshida, Tomoki Nakaya, Binbin Lu, Paul Harris (2023) **A linearization for stable and faster geographically weighted Poisson regression.** *International Journal of Geographical Information Science*, 37 (8), 1818–1839. DOI: 10.1080/13658816.2023.2209811 [LINK]
  17. Chaomin Chen, Hasi Bagan, Takahiro Yoshida (2023) **Multi-scale mapping of local climate zones in Tokyo using airborne LiDAR data, GIS vectors, and Sentinel-2 imagery.** *GIScience & Remote Sensing*, 60 (1), 2209970. DOI: 10.1080/15481603.2023.2209970 [LINK]
  18. Junya Yamasaki, Yasutaka Wakazuki, Satoru Iizuka, Takahiro Yoshida, Ryoichi Nitantai, Rikutarō Manabe, Akito Murayama (2023) **Diurnal change of summer temperature distribution in urban area based on SSP/RCP images under climate change: Using an analysis model compared with actual measurement results.** *Journal of the City Planning Institute of Japan*, 58 (3), 835–842. [in Japanese] DOI: 10.11361/journalcpj.58.835 [LINK]
  19. Mai Fukaya, Junya Yamasaki, Takahiro Yoshida, Ryoichi Nitantai, Rikutarō Manabe, Akito Murayama (2023) **Realities and prospects of area-based management activities aimed at solving social issues: Through a survey of advanced cases in Japan.** *Journal of the City Planning Institute of Japan*, 58 (3), 1320–1327. [in Japanese] DOI: 10.11361/journalcpj.58.1320 [LINK]
  20. Thiti Jittayasotorn, Muthiah Sadidah, Takahiro Yoshida, Takuro Kobashi (2023) **On the adoption of rooftop photovoltaics integrated with**

- electric vehicles toward sustainable Bangkok City, Thailand. *En-  
ergies*, 16 (7), 3011. DOI: 10.3390/en16073011 [LINK]
21. Junya Yamasaki, Wenchao Wu, Takahiro Yoshida, Ryoichi Nitnai, Rikutarō Manabe, Akito Murayama (2023) **Environmental impact assessment for future changes in land use composition under SSP scenarios in Japan: Assessment and issues based on the LCIA method LIME3**. *Environmental Science*, 36 (4), 135–144. [in Japanese] DOI: 10.11353/sesj.36.135 [LINK]
  22. Xinyan Hou, Xuan Xie, Hasi Bagan, Chaomin Chen, Qinxue Wang, Takahiro Yoshida (2023) **Exploring spatiotemporal variations in land surface temperature based on local climate zones in Shanghai from 2008 to 2020**. *Remote Sensing*, 15 (12), 3106. DOI: 10.3390/rs15123106 [LINK]
  23. Chaomin Chen, Hasi Bagan, Takahiro Yoshida, Habura Borjigin, Jun Gao (2022) **Quantitative analysis of the building-level relationship between building form and land surface temperature using airborne LiDAR and thermal infrared data**. *Urban Climate*, 45, 101248. DOI: 10.1016/j.uclim.2022.101248 [LINK]
  24. Joseph Emile Honour Percival, Narumasa Tsutsumida, Daisuke Murakami, Takahiro Yoshida, Tomoki Nakaya (2022) **Exploratory spatial data analysis with gwpcorMapper: An interactive mapping tool for geographically weighted correlation and partial correlation**. *Journal of Geovisualization and Spatial Analysis*, 6, 17. DOI: 10.1007/s41651-022-00111-3 [LINK]
  25. Junya Yamasaki, Masahiro Masubuchi, Yasutaka Wakazuki, Satoru Iizuka, Takahiro Yoshida, Ryoichi Nitnai, Rikutarō Manabe, Akito Murayama (2022) **Thermal environment simulation for future images based on SSP/RCP in urban center neighborhood under climate change: Focusing on the Nishiki 2 district, Nagoya city**. *Journal of the City Planning Institute of Japan*, 57 (3), 949–956. [in Japanese] DOI: 10.11361/journalcpij.57.949 [LINK]
  26. Tomoki Hosaka, Junya Yamasaki, Takahiro Yoshida, Ryoichi Nitnai, Rikutarō Manabe, Akito Murayama (2022) **Framework of climate change adaptation measures of urban planning-related sectors in UK and French municipalities: From the analysis of measures in 8 advanced Climate Change Action Plans**. *Journal of the City Planning Institute of Japan*, 57 (1), 138–150. [in Japanese] DOI: 10.11361/journalcpij.57.138 [LINK]
  27. Junya Yamasaki, Masahiro Masubuchi, Satoru Iizuka, Takahiro Yoshida, Ryoichi Nitnai, Rikutarō Manabe, Akito Murayama (2022) **Thermal environment simulation for urban center neighborhood by Using 3D city model of Project PLATEAU: Toward heat adaptation in community planning**. *Papers on Environmental Information Science*, 36, 238–243. [in Japanese] DOI: 10.11492/ceispapers.ceis36.0\_238 [LINK]
  28. Daisuke Murakami, Takahiro Yoshida, Yoshiki Yamagata (2021) **Gridded GDP projections compatible with the five SSPs (shared socioe-**

- conomic pathways). *Frontiers in Built Environment*, 7, 760306. DOI: 10.3389/fbuil.2021.760306 [LINK] [DATA] [SUPPORT SITE]
29. Soshiro Ogata, Misa Takegami, Taira Ozaki, Takahiro Nakashima, Daisuke Onozuka, Shunsuke Murata, Yuriko Nakaoku, Koyu Suzuki, Akihito Hagihara, Teruo Noguchi, Koji Iihara, Keiichi Kitazume, Tohru Morioka, Shin Yamazaki, Takahiro Yoshida, Yoshiki Yamagata, Kunihiro Nishimura (2021) **Heatstroke predictions by machine learning, weather information, and an all-population registry for 12-hour heatstroke alerts.** *Nature Communications*, 12, 4575. DOI: 10.1038/s41467-021-24823-0 [LINK]
  30. Meng Cai, Yuan Shi, Chao Ren, Takahiro Yoshida, Yoshiki Yamagata, Chao Ding, Nan Zhou (2021) **The need for urban form data in spatial modeling of urban carbon emissions in China: A critical review.** *Journal of Cleaner Production*, 319, 128792. DOI: 10.1016/j.jclepro.2021.128792 [LINK]
  31. Yasuo Takahashi, Hiroyuki Kubota, Sawako Shigeto, Takahiro Yoshida, Yoshiki Yamagata (2021) **Diverse values of urban-to-rural migration: A case study of Hokuto City, Japan.** *Journal of Rural Studies*, 87, 292–299. DOI: 10.1016/j.jrurstud.2021.09.013 [LINK]
  32. Mai Fukaya, Junya Yamasaki, Ryoichi Nitani, Rikutarō Manabe, Takahiro Yoshida, Akito Murayama (2021) **Long-term environmental changes in condominium complex and its surrounding neighborhoods: Toward the development of neighborhood sustainability assessment framework.** *Journal of the City Planning Institute of Japan*, 56 (3), 905–912. [in Japanese] DOI: 10.11361/journalcpj.56.905 [LINK]
  33. Narumasa Tsutsumida, Takahiro Yoshida, Daisuke Murakami, Tomoki Nakaya (2021) **A review on geographically weighted methods and their future directions.** *Theory and Applications of GIS*, 29 (1), 11–21. [in Japanese] DOI: 10.5638/thagis.29.11 [LINK]
  34. Takuro Kobashi, Peraphan Jittrapirom, Takahiro Yoshida, Yujiro Hirano, Yoshiki Yamagata (2021) **SolarEV City concept: Building the next urban power and mobility systems.** *Environmental Research Letters*, 16 (2), 024042. DOI: 10.1088/1748-9326/abd430 [LINK]
  35. Anna Ebata, Takahiro Yoshida, Kazuki Tamesue, Hajime Seya, Morito Tsutsumi (2021) **Exploratory spatial data analysis of *Furusato Nozei*: A tax donation program in Japan.** *Theory and Applications of GIS*, 29 (1), 1–10. [in Japanese] DOI: 10.5638/thagis.29.1 [LINK]
  36. Daisuke Murakami, Narumasa Tsutsumida, Takahiro Yoshida, Tomoki Nakaya, Binbin Lu (2021) **Scalable GWR: A linear-time algorithm for large-scale geographically weighted regression with polynomial kernels.** *Annals of the American Association of Geographers*, 111 (2), 459–480. DOI: 10.1080/24694452.2020.1774350 [LINK]
  37. Soowon Chang, Takahiro Yoshida, Daniel Castro-Lacouture, Yoshiki Yamagata (2021) **Block-level building transformation strategies for energy efficiency, thermal comfort and visibility using Bayesian**

- multilevel modeling.** *Journal of Architectural Engineering*, 27 (3), 05021008. DOI: 10.1061/(ASCE)AE.1943-5568.0000491 [LINK]
38. Yoshiki Yamagata, Takahiro Yoshida (2020) **A “Smart Lifestyle” for the re-design of the “After Corona” urban forms.** *Environment and Planning B: Urban Analytics and City Science*, 47 (7), 1146–1148. DOI: 10.1177/2399808320950043 [LINK]
  39. Takuro Kobashi, Takahiro Yoshida, Yoshiki Yamagata, Katsuhiko Naito, Stefan Pfenninger, Kelvin Say, Yasuhiro Takeda, Amanda Ahl, Masaru Yarime, Keishiro Hara (2020) **On the potential of “photovoltaics + electric vehicles” for deep decarbonization of Kyoto’s power systems: Techno-economic-social considerations.** *Applied Energy*, 275, 115419. DOI: 10.1016/j.apenergy.2020.115419 [LINK]
  40. Takuro Kobashi, Kelvin Say, Jiayang Wang, Masaru Yarime, Dong Wang, Takahiro Yoshida, Yoshiki Yamagata (2020) **Techno-economic assessment of photovoltaics plus electric vehicles towards household-sector decarbonization in Kyoto and Shenzhen by the year 2030.** *Journal of Cleaner Production*, 253, 119933. DOI: 10.1016/j.jclepro.2019.119933 [LINK]
  41. Takahiro Yoshida, Rim Er-rbib, Morito Tsutsumi (2019) **Which country epitomizes the world? A study from the perspective of demographic composition.** *Sustainability*, 11 (22), 6404. DOI: 10.3390/su11226404 [LINK]
  42. Michael B. Tobey, Robert B. Binder, Soowon Chang, Takahiro Yoshida, Yoshiki Yamagata, Perry P. J. Yang (2019) **Urban systems design: A conceptual framework for planning smart communities.** *Smart Cities*, 2 (4), 522–537. DOI: 10.3390/smartcities2040032 [LINK]
  43. Michael B. Tobey, Robert B. Binder, Takahiro Yoshida, Yoshiki Yamagata (2019) **Urban systems design case study: Tokyo’s Sumida ward.** *Smart Cities*, 2 (4), 453–470. DOI: 10.3390/smartcities2040028 [LINK]
  44. Yoshiki Yamagata, Takahiro Yoshida, Daisuke Murakami, Tomoko Matsui, Yuki Akiyama (2018) **Seasonal urban carbon emission estimation using spatial micro big data.** *Sustainability*, 10 (12), 4472. DOI: 10.3390/su10124472 [LINK]
  45. Takahiro Yoshida, Morito Tsutsumi (2018) **On the effects of spatial relationships in spatial compositional multivariate models.** *Letters in Spatial and Resource Sciences*, 11 (1), 57–70. DOI: 10.1007/s12076-017-0199-5 [LINK]
  46. Takahiro Yoshida, Morito Tsutsumi (2017) **The epitome of the future Japan from the perspective of demographic composition: Application of compositional data analysis to population studies.** *Theory and Applications of GIS*, 25 (2), 79–89. [in Japanese] DOI: 10.5638/thagis.25.79 [LINK]
  47. Daisuke Murakami, Takahiro Yoshida, Hajime Seya, Daniel A. Griffith, Yoshiki Yamagata (2017) **A Moran coefficient-based mixed effects approach to investigate spatially varying relationships.** *Spatial Statistics*, 19, 68–89. DOI: 10.1016/j.spasta.2016.12.001 [LINK]

48. Hajime Seya, Takahiro Yoshida, Morito Tsutsumi (2016) **Ex-post identification of geographical extent of benefited area by a transportation project: Functional data analysis method.** *Journal of Transport Geography*, 55, 1–10. DOI: 10.1016/j.jtrangeo.2016.07.004 [LINK]
49. Yoshiki Yamagata, Daisuke Murakami, Takahiro Yoshida, Hajime Seya, Sho Kuroda (2016) **Value of urban views in a bay city: Hedonic analysis with the spatial multilevel additive regression (SMAR) model.** *Landscape and Urban Planning*, 151, 89–102. DOI: 10.1016/j.landurbplan.2016.02.008 [LINK]

## Refereed proceedings papers

1. Ryuta Tsurumi, Asuka Tanzawa, Takahiro Yoshida (2026) **Identification of potential sites for district heating and cooling using observed operational carbon data.** *Energy Proceedings*, accepted. DOI: — [LINK]
  - This paper will be presented at the 18th International Conference on Applied Energy (ICAE2026), Jeju, South Korea, September 22–26, 2026.
2. Ryuta Tsurumi, Takahiro Yoshida (2026) **Spatiotemporal analysis of major CO<sub>2</sub> emission sources in the Japanese building sector.** E3S Web of Conferences, 716, 10013. DOI: 10.1051/e3sconf/202671610013 [LINK]
  - This paper was presented at the 12th International Conference on Indoor Air Quality, Ventilation & Energy Conservation in Buildings (IAQVEC 2026), Los Angeles, USA, May 18–22, 2026.
3. Qinghao Zeng, Ryan T. Nation, Hina I. Ahmed, Hsu-Chieh Ma, Kaiyu Zhou, Jiaqi Gu, Takahiro Yoshida, Akito Murayama, Perry Pei-Ju Yang (2025) **Enhancing Urban Heat Risk Resilience in Tokyo’s Nihonbashi through Urban Digital Twins of 4-Step Scenario Planning.** *The International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, XLVIII-4/W16-2025, 135–142. DOI: 10.5194/isprs-archives-xlvi-4-w16-2025-135-2025 [LINK]
  - This paper was presented at the 20th International 3D GeoInfo Conference 2025 and the 9th International Smart Data and Smart Cities Conference (3DGeoInfo2025 and SDSC2025), Kashiwa, Japan, September 2–5, 2025.
4. Abdulrahman H. Alorabi, Jingyuan Shen, Jalisa M. Smith, Max Hawkins, Kamyar Fatemifar, Takahiro Yoshida, Akito Murayama, Perry Pei-Ju Yang (2025) **A Data-driven Smart District Toward Net Zero Using Generative Design and Urban Digital Twins: A Use Case of Nihonbashi, Tokyo.** *The International Archives of the Photogrammetry,*

*Remote Sensing and Spatial Information Sciences*, XLVIII-4/W16-2025, 1–8. DOI: 10.5194/isprs-archives-xxviii-4-w16-2025-1-2025 [LINK]

- This paper was presented at the 20th International 3D GeoInfo Conference 2025 and the 9th International Smart Data and Smart Cities Conference (3DGeoInfo2025 and SDSC2025), Kashiwa, Japan, September 2–5, 2025.
5. Takahiro Yoshida, Daisuke Murakami, Ryuta Tsurumi, Hiroaki Onodera, Yujiro Hirano (2025) **Clustering Japanese municipalities from a perspective of energy demand sectors using compositional data analysis with a total.** *Energy Proceedings*, 59, 11896. DOI: 10.46855/energy-proceedings-11896 [LINK]
    - This paper was presented at the Applied Energy Symposium and Forum: Low carbon cities and urban energy systems (CUE2025), Kitakyushu, Japan, July 18–22, 2025.
  6. Daisuke Murakami, Takahiro Yoshida, Naoki Hiramoto (2025) **Projecting building-level gross product under Shared Socioeconomic Pathways.** *Energy Proceedings*, 60, 11925. DOI: 10.46855/energy-proceedings-11925 [LINK]
    - This paper was presented at the Applied Energy Symposium and Forum: Low carbon cities and urban energy systems (CUE2025), Kitakyushu, Japan, July 18–22, 2025.
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## Book chapters / sections

1. Takahiro Yoshida (2025) **Urban Carbon Mapping.** In Hideki Koizumi, Hikaru Kobayashi, Fumihiko Seta, Akito Murayama (eds.) *Urban Planning Towards Carbon Neutrality*, 39–44, Gakugei Publishing. [in Japanese] [LINK]
2. Takahiro Yoshida (2025) **Quality of life.** In Peter Nijkamp, Karima Kourtit, Kingsley Haynes, Zeynep Elburz (eds.) *The Thematic Encyclopedia of Regional Science*, 580–581, Edward Elgar Publishing. DOI: 10.4337/9781800379282 [LINK]

3. Yoshiki Yamagata, Daisuke Murakami, Hajime Seya, Takahiro Yoshida (2024) **High-resolution remote sensing and visibility analysis method for smart environment design.** In Prasad S. Thenkabail (ed.) *Remote Sensing Handbook, Volume VI: Droughts, Disaster, Pollution, and Urban Mapping (the 2nd edition)*, 362–378, CRC Press. DOI: 10.1201/9781003541417-17 [LINK]
4. Yoshiki Yamagata, Takahiro Yoshida (2021) **Urban carbon mapping: Spatio-temporal detail visualization for CO2 emissions.** In Takuro Kobashi (ed.) *Urban Decarbonization*, 65–72, TAIGA Publishing. [in Japanese] [LINK]
5. Kazuto Sumita, Motoi Okamoto, Masamune Iwasawa, Yanchun Jin, Yoichi Mizumura, Takahiro Yoshida (2020) **Introduction to Economic Data Analysis with Excel and R**, Ohmsha. [in Japanese] [LINK]
6. Yoshiki Yamagata, Takahiro Yoshida (2020) **Kyojima smart community studio: Urban systems design.** In Thomas Fisher, Brian Orland, Carl Steinitz (eds.) *The International Geodesign Collaboration: Changing Geography by Design*, 132–133, Esri Press. [LINK]
7. Yoshiki Yamagata, Takahiro Yoshida, Perry P. J. Yang, Helen Chen, Daisuke Murakami, Leena Ilmola (2020) **Measuring quality of walkable urban environment through experiential modeling.** In Yoshiki Yamagata, Perry P. J. Yang (eds.) *Urban Systems Design: Creating Sustainable Smart Cities in the Internet of Things Era*, 373–392, Elsevier. DOI: 10.1016/B978-0-12-816055-8.00012-9 [LINK]
8. Takuro Kobashi, Yoshiki Yamagata, Takahiro Yoshida, Soowon Chang, Yasunori Mochizuki, Amanda Ahl, Jelena Aleksejeva (2020) **Smart city and ICT infrastructure with vehicle to X applications toward urban decarbonization.** In Yoshiki Yamagata, Perry P. J. Yang (eds.) *Urban Systems Design: Creating Sustainable Smart Cities in the Internet of Things Era*, 289–333, Elsevier. DOI: 10.1016/B978-0-12-816055-8.00009-9 [LINK]
9. Yoshiki Yamagata, Takahiro Yoshida, Soowon Chang, Peraphan Jittrapirom, Sylvia Coleman, John Robinson, Roger Cremades, Dirk Neumann (2020) **Case studies towards smart communities.** In Yoshiki Yamagata, Perry P. J. Yang (eds.) *Urban Systems Design: Creating Sustainable Smart Cities in the Internet of Things Era*, 257–288, Elsevier. DOI: 10.1016/B978-0-12-816055-8.00008-7 [LINK]
10. Takahiro Yoshida, Yoshiki Yamagata, Soowon Chang, Vincent de Gooyert, Hajime Seya, Daisuke Murakami, Peraphan Jittrapirom, Gerasimos Voulgaris (2020) **Spatial modelling and design of smart communities.** In Yoshiki Yamagata, Perry P. J. Yang (eds.) *Urban Systems Design: Creating Sustainable Smart Cities in the Internet of Things Era*, 199–255, Elsevier. DOI: 10.1016/B978-0-12-816055-8.00007-5 [LINK]
11. Yoshiki Yamagata, Perry P. J. Yang, Soowon Chang, Michael B. Tobey, Robert B. Binder, Pieter J. Fourie, Peraphan Jittrapirom, Takuro Kobashi, Takahiro Yoshida, Jelena Aleksejeva (2020) **Urban systems and the role of big data.** In Yoshiki Yamagata, Perry P. J. Yang (eds.)

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  13. Takahiro Yoshida, Yoshiki Yamagata (2019) **Quasi real time energy use estimation using Google’s popular time data**. In Yoshiki Yamagata, Hajime Seya (eds.) *Spatial Analysis Using Big Data: Methods and Urban Applications*, 271–280, Academic Press. DOI: 10.1016/B978-0-12-813127-5.00011-4 [LINK]
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  15. Takahiro Yoshida, Daisuke Murakami (2019) **Implementation with R language**. In Yoshiki Yamagata, Hajime Seya (eds.) *Spatial Analysis Using Big Data: Methods and Urban Applications*, 181–223, Academic Press. DOI: 10.1016/B978-0-12-813127-5.00007-2 [LINK]
  16. Hajime Seya, Takahiro Yoshida, Yoshiki Yamagata (2019) **Spatial econometric models**. In Yoshiki Yamagata, Hajime Seya (eds.) *Spatial Analysis Using Big Data: Methods and Urban Applications*, 113–158, Academic Press. DOI: 10.1016/B978-0-12-813127-5.00005-9 [LINK]
  17. Takahiro Yoshida, Morito Tsutsumi (2017) **Cluster analysis**. In Committee of Infrastructure Planning and Management (eds.) *Handbook of Infrastructure Planning and Management*, 118–120, Corona Publishing. [in Japanese] [LINK]

## Book translation

1. Giuseppe Arbia (2014) *A Primer for Spatial Econometrics: With Applications in R*. Palgrave Macmillan. Translated by: Morito Tsutsumi (supervision), Takahiro Yoshida, Kazuki Tamesue, Keisuke Takano, Masato Tomonari (2016) . Keiso shobo. [in Japanese] [LINK]

# Honors

## Awards

- **Poster Session Award**, GIS Association of Japan, November 2025. (with Ikuho Yamada)
- **Excellent Joint Research Presentation Award**, CSIS DAYS 2024 (Center for Spatial Information Science, The University of Tokyo), November 2024. (with Taku Murakami and Narumasa Tsutsumida)
- **Poster Session Award**, GIS Association of Japan, October 2023. (with Daisuke Murakami and Hajime Seya)
- **Excellent Joint Research Presentation Award**, CSIS DAYS 2022 (Center for Spatial Information Science, The University of Tokyo), November 2022. (with Narumasa Tsutsumida, Joseph Percival, Daisuke Murakami, and Tomoki Nakaya)
- **Map Gallery Award (the first place)**, ESRI Japan, Corp., May 2022. (with Kazushi Matsuo, Yujiro Hirano, and Akito Murayama)
- **Excellent Research Presentation Award**, CSIS DAYS 2018 (Center for Spatial Information Science, The University of Tokyo), November 2018. (with Yoshiaki Yamagata, Daisuke Murakami, and Tomoko Matsui)
- **GISA Award: Research Initiative Category**, GIS Association of Japan, October 2018. [LINK]
- **Poster Session Award**, GIS Association of Japan, October 2018. (with Anna Ebata, Kazuki Tamesue, Hajime Seya, and Morito Tsutsumi)
- **Map Gallery Award (the second place)**, ESRI Japan, Corp., May 2018. (with Anna Ebata)
- **Dean's Award**, Graduate School of Systems and Information Engineering, University of Tsukuba, March 2018.
- **Young Scholar Award**, Environmental Systems Research Institute (ESRI), Inc., July 2017. [LINK]
- **President's Award**, University of Tsukuba, March 2017. (with Marisa Yamaguchi and Rim Er-Rbib)
- **Poster Session Award**, GIS Association of Japan, October 2016. (with Morito Tsutsumi)
- **Map Gallery Award (the fourth place)**, ESRI Japan, Corp., May

2016. (with Shuhei Adachi, Yoi Kana, and Naoya Yanagisawa)

- **Dean's Award**, Graduate School of Systems and Information Engineering, University of Tsukuba, March 2015.
- **S-PLUS Student Research Encouragement Award (the first place)**, NTT DATA Mathematical Systems, Inc., November 2014.
- **Excellent Presentation Award**, GIS Association of Japan, November 2014.

## Research grants

### Principal Investigator

- June 2025 – March 2026
  - Electricity Data Utilization Support Project, Green Investment Promotion Organization. Project Number: S-010. Budget Amount: JPY 55,037,180.
- April 2025 – March 2029
  - Grant-in-Aid for Scientific Research (B), Japan Society for the Promotion of Science. Project Number: 25K00624. Budget Amount: JPY 18,460,000.
- June 2024 – March 2025
  - Collaboration Program at Joint Support-Center for Data Science Research, Research Organization of Information and Systems. Project Number: 055RP2024. Budget Amount: JPY 712,000.
- April 2021 – March 2025
  - Grant-in-Aid for Early-Career Scientists, Japan Society for the Promotion of Science. Project Number: 21K13153. Budget Amount: JPY 4,680,000.
- April 2016 – March 2018
  - Grant-in-Aid for JSPS Fellows, Japan Society for the Promotion of Science. Project Number: 16J02219. Budget Amount: JPY 2,300,000.

### Co-Investigator

- April 2024 – March 2028
  - Grant-in-Aid for Scientific Research (B) (PI: Ryo Inoue), Japan Society for the Promotion of Science. Project Number: 24K00997.

- April 2024 – March 2028
  - Grant-in-Aid for Scientific Research (B) (PI: Wataru Nakanishi), Japan Society for the Promotion of Science. Project Number: 24K01004
- April 2024 – March 2027
  - Environment Research and Technology Development Fund (1RA-2402) (PI: Daisuke Murakami), Environmental Restoration and Conservation Agency. Project Number: JPMEERF20241RA2
- July 2023 – March 2028
  - Cross-ministerial Strategic Innovation Promotion Program (SIP) (PIs: Yumiko Iwafune (July 2023 – March 2026) and Yoshiaki Yamagata (April 2026 – March 2028)), Council for Science, Technology and Innovation of Cabinet Office. Project Number: JPJ012207
- April 2023 – March 2028
  - Grant-in-Aid for Scientific Research (B) (PI: Akito Murayama), Japan Society for the Promotion of Science. Project Number: 23K26272 (23H01578)
- April 2023 – March 2026
  - Grant-in-Aid for Scientific Research (B) (PI: Yujiro Hirano), Japan Society for the Promotion of Science. Project Number: 23K26240 (23H01546)
- April 2023 – March 2028
  - Environment Research and Technology Development Fund (S-21-5) (PI: Takehito Yoshida), Environmental Restoration and Conservation Agency. Project Number: JPMEERF23S12150
- April 2023 – March 2026
  - Environment Research and Technology Development Fund (SII-11-1) (PI: Taikan Oki), Environmental Restoration and Conservation Agency. Project Number: JPMEERF23S21110
- June 2021 – March 2024
  - Collaboration Program at Joint Support-Center for Data Science Research (PI: Narumasa Tsutsumida), Research Organization of Information and Systems. Project Numbers: 005RP2021, 006RP2022, and 010RP2023
- April 2021 – March 2023
  - Grant-in-Aid for Scientific Research (A) (PI: Morito Tsutsumi), Japan Society for the Promotion of Science. Project Number: 18H03628

- April 2019 – March 2023
  - Grant-in-Aid for Challenging Research (Exploratory) (PI: Morito Tsutsumi), Japan Society for the Promotion of Science. Project Number: 17K18554
- October 2018 – March 2022
  - Topic-Setting Program to Advance Cutting-Edge Humanities and Social Sciences Research (Responding to Real Society) (PI: Yasuo Takahashi), Japan Society for the Promotion of Science.
- June 2018 – March 2021
  - Collaboration Program at Joint Support-Center for Data Science Research (PI: Narumasa Tsutsumida), Research Organization of Information and Systems. Project Numbers: 006RP2018, 004RP2019, and 003RP2020

## Research Collaborator

- November 2020 – March 2023
  - JST-Mirai Program (PI: Norio Ohmagari), Japan Science and Technology Agency. Project Number: JPM.JMI20B2
- April 2016 – March 2019
  - Grant-in-Aid for Scientific Research (B) (PI: Yoshiki Yamagata), Japan Society for the Promotion of Science. Project Number: 16H02910
- April 2015 – March 2018
  - Grant-in-Aid for Scientific Research (B) (PI: Morito Tsutsumi), Japan Society for the Promotion of Science. Project Number: 15H04054

# Notes: R and Python

- Under construction ...
  - This page will summarize how to run spatial analysis and spatial data analysis using R or Python.
  - Currently, some tech-blogs in Japanese are being posted to Qiita. Recent posts are:
    - \* [Microsoft Bing Maps] R
    - \* QGIS PLATEAU LOD2
    - \* m
    - \* MODIS R LST
    - \* [R] gdb GIS sf
- Translation in progress ...
  - Robin Lovelace, Jakub Nowosad, Jannes Muenchow (2019) ***Geocomputation with R***. Chapman and Hall/CRC Press.
  - Translators: Shinya Uryu, Narumasa Tsutsumida, Takahiro Yoshida.